

O–A–V Annotations for Eight Representative PriorPair Examples

Demo 1: Coherence Violation

Positive video

Observation: A woman wearing sunglasses, a grey hoodie, a black vest, and leggings walks down a series of wooden steps on a dirt trail surrounded by dry vegetation and trees. She carries a small, light-colored dog inside a bright blue sling bag strapped across her chest. As she descends the path, she moves steadily towards the camera, bringing the dog in the carrier into close view before passing the lens.

Attribution: The sequence of events perfectly adheres to the linearity of time and the micro-temporal continuity of physical actions. The continuous movement of the woman descending the steps, gradually coming closer to the camera, and eventually passing the lens unfolds in a natural, irreversible chronological sequence without spatial or temporal interruptions.

Verdict: The video is physically valid.

Negative video

Observation: A woman wearing sunglasses, a grey hoodie, a black vest, and leggings is seen in close view, carrying a small, light-colored dog in a bright blue sling bag while moving towards the camera on a dirt trail. Suddenly, her position resets back to the top of a series of wooden steps surrounded by dry vegetation. An instant later, she is seen passing the camera lens, before abruptly jumping backward again to a mid-descent state walking down the path.

Attribution: This disjointed jumping of states violates the fundamental linearity of time and the macro-causality of motion, explicitly constituting a Coherence Violation. The continuous progression of the woman’s walking action is broken at a micro-temporal level because the chronological order of frames is scrambled. This results in the subject abruptly teleporting between different stages of a single continuous descent (start, end, middle) without any fluid or logical transition.

Verdict: The video is physically invalid.

Demo 2: Causal Reversal

Positive video

Observation: On a wooden surface, two black and blue clamps grip the opposing edges of a green rectangular sheet. The clamps twist and pull outwards. Consequently, the green sheet warps and buckles in the center, subsequently tearing in half and separating into two distinct pieces that are pulled apart by the clamps.

Attribution: This sequence of events perfectly adheres to macroscopic causality, where the physical cause logically and chronologically precedes the effect. The mechanical tension and pulling force applied by the clamps realistically cause the subsequent deformation and structural failure of the material.

Verdict: The video is physically valid.

Negative video

Observation: On a wooden surface, a green rectangular sheet is seen warping and buckling in the center, subsequently tearing in half and separating into two distinct pieces. Abruptly, the scene cuts to two black and blue clamps gripping the opposing edges of the intact green sheet, twisting and pulling outwards.

Attribution: This sequence violates the Linearity and Causality of Time. Presenting the macroscopic effect (the sheet warping and tearing) before its physical cause (the clamps gripping and pulling the intact sheet) constitutes a Causal Reversal, contradicting the fundamental arrow of time.

Verdict: The video is physically invalid.

Demo 3: Existence Violation

Positive video

Observation: A suspended cylindrical wicker basket descends onto a wooden table, completely covering a static orange tennis ball. After a brief pause where the ball is concealed inside the basket, the basket is lifted vertically, revealing the tennis ball remaining untouched in its original position on the table.

Attribution: This phenomenon adheres to the fundamental principle of Conservation of Matter and Morphological Stability. The basket merely acts as a temporary visual cover. Without any explicit physical interaction or external force extracting the ball from beneath the basket, the object maintains its continuous existence and spatial position.

Verdict: The video is physically valid.

Negative video

Observation: A suspended cylindrical wicker basket descends onto a wooden table, completely covering a static orange tennis ball. After a brief pause, the basket is lifted vertically, but the tennis ball is no longer present on the table, and the area under the basket is revealed to be empty wood.

Attribution: This scenario violates the core physical principle of Conservation of Matter and Morphological Stability, specifically demonstrating an Existence Violation (Disappearance After Uncovering). Since the basket simply covered the tennis ball without any mechanism to remove or alter it, the object should remain consistently in its state and location; its spontaneous disappearance into thin air is physically impossible.

Verdict: The video is physically invalid.

Demo 4: Identity Violation

Positive video

Observation: A black and white speckled dog with a blue tag on its collar trots towards the camera on a grassy park lawn, with large trees and parked cars visible in the background. As the dog approaches, it lowers its head to sniff the ground while continuing its forward movement.

Attribution: This event adheres to the physical principle of Conservation of Identity and Intrinsic Properties. The dog consistently maintains its fundamental biological characteristics, form, and species throughout its continuous physical movement, with no unnatural mutations occurring.

Verdict: The video is physically valid.

Negative video

Observation: A black and white speckled dog with a blue tag on its collar trots towards the camera on a grassy park lawn. Mid-stride, as the dog lowers its head to sniff the ground, it seamlessly and instantly morphs into a sheep while continuing the exact same walking trajectory and forward cadence.

Attribution: This event violates the principle of Conservation of Identity and Intrinsic Properties, specifically demonstrating an Identity Violation via Spontaneous Transformation (Morphology/Species Mutation). The dog's intrinsic properties undergo a sudden, physically impossible mutation into a completely different species without any corresponding causal interaction or realistic transformative physical process.

Verdict: The video is physically invalid.

Demo 5: Dynamic Violation

Positive video

Observation: A yellow rubber duck with an orange beak sits completely stationary on a light brown wooden table. The background is a plain, light grey wall. The lighting casts a soft shadow to the right of the duck, and there is no movement from the object or the camera throughout the sequence.

Attribution: This scenario perfectly adheres to the principles of classical mechanics and the causal consistency of physical interactions. Specifically, it illustrates Newton's first law: the table's support force balances the duck's weight, so the net external force is zero and the duck remains at rest.

Verdict: The video is physically valid.

Negative video

Observation: A yellow rubber duck with an orange beak sits on a light brown wooden table against a plain, light grey wall with a soft shadow cast to its right. Without any visible external contact, wind, or tilting of the table, the rubber duck suddenly begins to slide smoothly across the wooden surface to the right. It accelerates from a standstill while maintaining its exact static pose throughout its traversal across the table.

Attribution: This phenomenon violates the 'Causal Consistency of Physical Interactions', specifically representing a 'Phantom Force (Causality without Contact)' dynamic violation. According to Newton's laws of motion, an object can accelerate only if a nonzero net external force acts upon it. The depicted spontaneous acceleration, without a corresponding external driving mechanism, violates this requirement.

Verdict: The video is physically invalid.

Demo 6: Constraint Violation

Positive video

Observation: A campfire constructed of logs and a stone ring burns brightly on a pebble beach. In the background, gentle ocean waves roll towards the shore near a large white cliff face under a cloudy twilight sky.

Attribution: This scene adheres to the principles of thermodynamics and natural environmental physics. The fire is fueled by the logs and oxygen in the open air, burning steadily within the safety of the stone ring, while the ocean waves remain at a distance in the background, causing no interference with the combustion process.

Verdict: The video is physically valid.

Negative video

Observation: A campfire constructed of logs and a stone ring is situated on a pebble beach near a large white cliff. Ocean waves surge forward and completely engulf the campfire. Once fully submerged, the bright orange flames continue to burn and flicker rhythmically underwater without being extinguished and without generating any steam.

Attribution: This sequence violates the 'Causal Consistency of Physical Interactions', specifically exhibiting a 'Thermodynamic & Environmental Disconnect'. A standard wood-fueled campfire requires gaseous oxygen to sustain combustion. When engulfed by water, the rapid loss of heat and oxygen should immediately extinguish the fire and produce steam. The flames continuing to burn steadily underwater blatantly contradicts the thermodynamic interactions and material constraints between liquid water and an open flame.

Verdict: The video is physically invalid.

Demo 7: Near-Miss

Target event: The player's foot contacts the soccer ball.

Positive video

Observation: A soccer ball rests on green artificial turf against a dark background. A player wearing white socks, blue shorts, and cleats approaches from the left, swings their right leg, and brings their foot into direct physical contact with the ball. Upon this impact, the ball is launched upward and forward into the darkness, while the player continues running through the frame.

Attribution: The objects meet at the same location with no visible gap, establishing direct contact.

Verdict: The target event occurred.

Negative video

Observation: A soccer ball rests on green artificial turf against a dark background. A player in white socks, blue shorts, and cleats approaches from the left and swings the right leg. The foot passes beside the ball with a visible gap. The soccer ball remains stationary as the player continues through the frame.

Attribution: A visible gap persists at the closest approach, so the objects never make contact.

Verdict: The target event did not occur.

Demo 8: Consequence Arrest

Target event: The yellow mug breaks.

Positive video

Observation: A yellow ceramic mug drops vertically from a blue hanging clip against a white background. It falls and strikes a concrete block resting on a wooden deck. Upon contact, the mug breaks into multiple shards that scatter across the floor.

Attribution: The brittle ceramic mug strikes the concrete block with enough contact stress to exceed its structural threshold, so it fractures and scatters into multiple pieces.

Verdict: The target event occurred.

Negative video

Observation: A yellow mug-shaped vessel drops onto the concrete block, topples off it, and settles on the wooden deck as one intact object without visible cracks or fragments.

Attribution: The mug-shaped vessel is made of impact-resistant material that withstands the drop.

Verdict: The target event did not occur.